Melissa (Millie) Chapman

melissa.chapman@usys.ethz.ch
http://milliechapman.info
milliechapman

Education

2018 – 2023 **Ph.D., UC Berkeley** Environmental Science, Policy, and Management.
Dissertation: *From individual decisions to international agreements: Addressing biodiversity loss in an age of algorithms*

2010 – 2014 **B.Sc. Yale University** Ecology and evolutionary biology.
Thesis: *Assessing patterns of malaria risk: Environmental and social determinants of endemicity across Burkina Faso and Kenya.*

Affiliations and Appointments

2025 – present **ETH Zürich** — Tenure Track Assistant Professor of Environmental Policy, in the Department of Environmental Systems Science. *Faculty member of:* Einstein School of Public Policy (Environmental Cluster Co-Lead); Center for Comparative and International Studies; ETH AI Center; Zurich–Basel Plant Sciences Center

Source Cooperative Science and Technology Advisory Board

IPBES Expert for the Spatial Planning and Connectivity Assessment

2024 – 2025 **Google Research** Visiting Faculty Researcher

2023 – 2025 **National Center for Ecological Analysis and Synthesis (NCEAS)** Director's Postdoctoral Fellow

Climate Change AI (CCAI) Core team

2022 – 2023 **International Institute of Applied Systems Analysis (IIASA)** Visiting Research Scholar | Visiting PhD Student

2021 – 2023 **Data Intensive Social Science Lab** Teaching Fellow

2020 – 2023 **Resources Legacy Fund** Lead scientific writer for California's Pathways to 30x30 Initiative

2015 – 2018 **Woodwell Climate Research Center** Research Assistant II

Publications

[Google Scholar] | **[ORCID]**

1. Ellis-Soto, D., & **Chapman**, **M.** (2026). Historic residential segregation impacts biodiversity data availability disparately across the tree of life [Published September 14, 2026]. *Proceedings of the National Academy of Sciences*, *123*(38), Article e2416889122. https://doi.org/10.1073/pnas.2416889122

2. Hulkunda, N., Swanson, A., Dehmler-Buckley, O., Burkart, K., **Chapman**, **M.**, & Beery, S. (2026). Models and metrics aren't enough: How data marketplaces can help close the biodiversity information gap [In press]. *Proceedings of the National Academy of Sciences.*

3. Neubert, S. et al. (2026). Advancing multi-objective optimization in spatial planning [In press]. *Journal of Applied Ecology.*

4. Perry, N., Schweier, J., Bont, L. G., Kushwaha, S. K. P., Peltola, H., Eyvindson, K., Astrup, R., **Chapman**, **M.**, & Blattert, C. (2026). Towards precision forestry: A systematic review of optimisation methods for individual-tree decisions in forest management. *Trees, Forests and People*, 101226.

5. **Chapman**, **M.**, Jung, M., Boettiger, C., Ringwald, L., Leclère, D., Gusti, M., Augustynczik, A., & Visconti, P. (2025). Meeting European biodiversity targets under future land-use demands. *Nature Ecology and Evolution*. https://doi.org/10.1038/s41559-025-02671-1

6. Faxon, H., & **Chapman**, **M.** (2025). Beyond spatial bias: Understanding the colonial legacies and contemporary social forces shaping biodiversity data. *Environmental Research Letters*. https://doi.org/10.1088/1748-9326/add6b6

7. Harrell, L., Kaeser-Chen, C., Ayan, B. K., Anderson, K., Conserva, M., Kleeman, E., Neumann, M., Overlan, M., **Chapman**, **M.**, & Purves, D. (2025). Heterogenous graph neural networks for species distribution modeling. *International Conference on Learning Representations (ICLR)*. https://doi.org/10.48550/arXiv.2503.11900

8. Johnson, J. A., Chaplin-Kramer, R., **Chapman**, **M.**, Polasky, S., & Williams, B. (2025). Earth-economy modeling: Advances in linking economic and ecosystem models. *Annual Review of Resource Economics*. https://doi.org/10.1146/annurev-resource-013024-033043

9. Oestreich, W., Czapanskiy, M., Katija, K., Record, N., & **Chapman**, **M.** (2025). Collective science to inform global ocean protections. *Ecology Letters*. https://doi.org/10.22541/au.174282888.81070415/v1

10. Scoville, C., Amironesei, R., Xu, L., **Chapman**, **M.**, Record, N., & Boettiger, C. (2025). From maps to models: Participation and contestability in the dynamic management of natural resources. *Geo: Geography and Environment*. https://doi.org/10.1002/geo2.70028

11. **Chapman**, **M.**, Goldstein, B., Schell, C., Brashares, J. S., Carter, N. H., Ellis-Soto, D., Faxon, H. O., Goldstein, J. E., Halpern, B. S., Longdon, J., Norman, K. E., O'Rourke, D., Scoville, C., Xu, L., & Boettiger, C. (2024). Biodiversity monitoring for a just planetary future. *Science*. https://doi.org/10.1126/science.adh8874

12. Ellis-Soto, D., **Chapman**, **M.**, & Koltz, A. (2024). Addressing data disparities is critical for biodiversity assessments. *Trends in Ecology and Evolution*. https://doi.org/10.1016/j.tree.2024.10.005

13. Jung, M., Alagador, D. A., **Chapman**, **M.**, Hermoso, V., Kujala, H., O'Connor, L., Schinegger, R., Verburg, P., & Visconti, P. (2024). An assessment of the state of conservation planning in Europe. *Philosophical Transactions of the Royal Society B*. https://doi.org/0.1098/rstb.2023.0015

14. Oestreich, W., Mckenna, M., Go, M., **Chapman**, **M.**, & Oliver, R. (2024). Listening to animal behavior to understand changing ecosystems. *Trends in Ecology and Evolution*. https://doi.org/10.1016/j.tree.2024.06.007

15. Oliver, R., **Chapman**, **M.**, Ellis-Soto, D., Brum-Bastos, V., Cagnacci, F., Long, J., Loretto, M.-C., Patchett, R., & Rutz, C. (2024). Access to human mobility data is essential for building a sustainable future. *Cell Reports Sustainability*. https://doi.org/10.1016/j.crsus.2024.10007

16. Oliver, R., **Chapman**, **M.**, Emery, N., Gillespie, L., Gownaris, N., Leiker, S., Nisi, A., Ayers, D., Breckheimer, I., Blondin, H., Hoffman, A., Pagniello, C., Raisle, M., & Zimmerman, N. (2024). Opening a conversation on responsible environmental data science in the age of generative AI. *Environmental Data Science*. https://doi.org/10.1017/eds.2024.12

17. Sprenkle-Hyppolite, S., Griscom, B., Griffey, V., Munshi, E., & **Chapman**, **M.** (2024). Expert-informed leakage-free global carbon dioxide removal potential of trees in agriculture. *Carbon Balance and Management*. https://doi.org/10.1186/s13021-024-00268-y

18. **Chapman**, **M.**, Boettiger, C., & Brashares, J. S. (2023). Leveraging private lands to meet 2030 biodiversity targets in the United States. *Conservation Science and Practice*. https://doi.org/0.1111/csp2.12897

19. **Chapman**, **M.**, Xu, L., Lapeyrolerie, M., & Boettiger, C. (2023). Bridging adaptive management and reinforcement learning for more robust decisions. *Philosophical Transactions of the Royal Society B*. https://doi.org/10.1098/rstb.2022.0195

20. Hasting, Z., **Chapman**, **M.**, Ocloo, X., Stenger, K., & Hunt, L. (2023). Trends in agroforestry research over four decades *co-first author. *Elementa: Science of the Anthropocene*. https://doi.org/10.1525/elementa.2022.0015

21. Kurz, D., Middleton, A. D., **Chapman**, **M.**, Van Houtan, K. S., Wilkinson, C., Withey, L., & Brashares, J. (2023). Including rural america in academic conservation science. *Frontiers in Conservation Science*. https://doi.org/10.3389/fcosc.2023.1227227/full

22. Moravek, J., Andrews, L. R., Serota, M. W., Dorcy, J. A., **Chapman**, **M.**, Wilkinson, C. E., Parker-Shames, P., Van Scoyoc, A., Verta, G., & Brashares, J. S. (2023). Centering 30x30 conservation initiatives on freshwater ecosystems. *Frontiers in Ecology and the Environment*. https://doi.org/10.1002/fee.2573

23. Scoville, C., Faxon, H., **Chapman**, **M.**, & et al. (2023). Environment, society and machine learning. *Handbook on the Sociology of Machine Learning*. https://doi.org/10.1093/oxfordhb/9780197653609.013.8

24. Xu, L., Rolf, E., Beery, S., Bennett, J. R., Berger-Wolf, T., Birch, T., Bondi-Kelly, E., Brashares, J., **Chapman**, **M.**, Corso, A. et al. (2023). Reflections from the workshop on AI-assisted decision making for conservation. *arXiv*. https://doi.org/10.48550/arXiv.2307.08774

25. Ellis-Soto, D., **Chapman**, **M.**, & Locke, D. (2023). Uneven biodiversity sampling across redlined urban areas in the united states. *Nature Human Behavior*. https://doi.org/10.1038/s41562-023-01688-5

26. Montealegre-Mora, F., Laperolerie, M., **Chapman**, **M.**, Keller, A., & Boettiger, C. (2023). Pretty darn good control: When are approximate solutions better than approximate models? *Bulletin of Mathematical Biology*. https://doi.org/10.1007/s11538-023-01198-5

27. Calhoun, K. L., **Chapman**, **M.**, Tubbesing, C., McInturff, A., Gaynor, K. M., Van Scoyoc, A., Wilkinson, C. E., Parker-Shames, P., Kurz, D., & Brashares, J. (2022). Spatial overlap of wildfire and biodiversity in california highlights gap in non-conifer fire research and management. *Diversity and Distributions*, *28*(3), 529–541. https://doi.org/10.1111/ddi.13394

28. **Chapman**, **M.**, Wiltshire, S., Baur, P., Bowles, T., Carlisle, L., Castillo, F., Esquivel, K., Gennet, S., Iles, A., Karp, D. et al. (2022). Social-ecological feedbacks drive tipping points in farming system diversification. *One Earth*, *5*(3), 283–292. https://doi.org/10.1016/j.oneear.2022.02.007

29. Dowd, S., **Chapman**, **M.**, Koehn, L. E., & Hoagland, P. (2022). The economic tradeoffs and ecological impacts associated with a potential mesopelagic fishery in the california current. *Ecological Applications*, e2578. https://doi.org/10.1002/eap.2578

30 Lapeyrolerie, M., **Chapman**, **M.**, Norman, K. E., & Boettiger, C. (2022). Deep reinforcement learning for conservation decisions. *Methods in Ecology and Evolution*. https://doi.org/10.48550/arXiv.2106.08272

31 Estein, C., **Chapman**, **M.**, Schell, C., Lowy, N., & Gerson, J. (2022). Demystifying the graduate school application process. *Bulletin of the Ecological Society of America*. https://doi.org/10.1002/bes2.2029

32 **Chapman**, **M.**, Oestreich, W. K., Frawley, T. H., Boettiger, C., Diver, S., Santos, B. S., Scoville, C., Armstrong, K., Blondin, H., Chand, K. et al. (2021). Promoting equity in the use of algorithms for high-seas conservation. *One Earth*, *4*(6), 790–794. https://doi.org/10.1016/j.oneear.2021.05.011

33 **Chapman**, **M.**, Scoville, C., Lapeyrolerie, M., & Boettiger, C. (2021). Power and accountability in reinforcement learning applications to environmental policy. *The Thirty-Sixth Annual Conference on Neural Information Processing Systems (NeurIPS 2021)*. https://doi.org/10.48550/arXiv.2205.10911

34 Kitzes, J., Blake, R., Bombaci, S., **Chapman**, **M.**, Duran, S. M., Huang, T., Joseph, M. B., Lapp, S., Marconi, S., Oestreich, W. K. et al. (2021). Expanding neon biodiversity surveys with new instrumentation and machine learning approaches. *Ecosphere*, *12*(11), e03795. https://doi.org/10.1002/ecs2.3795

35 Nagy, R. C., Balch, J. K., Bissell, E. K., Cattau, M. E., Glenn, N. F., Halpern, B. S., Ilangakoon, N., Johnson, B., Joseph, M. B., **Chapman**, **M.** et al. (2021). Harnessing the neon data revolution to advance open environmental science with a diverse and data-capable community. *Ecosphere*, *12*(12), e03833. https://doi.org/10.1002/ecs2.3833

36 Ordway, E. M., Elmore, A. J., Kolstoe, S., Quinn, J. E., Swanwick, R., Cattau, M., Taillie, D., Guinn, S. M., Chadwick, K. D., **Chapman**, **M.** et al. (2021). Leveraging the neon airborne observation platform for socio-environmental systems research. *Ecosphere*, *12*(6), e03640. https://doi.org/10.1002/ecs2.3640

37 Roe, S., Streck, C., Beach, R., Busch, J., **Chapman**, **M.**, Daioglou, V., Deppermann, A., Doelman, J., Emmet-Booth, J., Engelmann, J. et al. (2021). Land-based measures to mitigate climate change: Potential and feasibility by country. *Global Change Biology*, *27*(23), 6025–6058. https://doi.org/10.1111/gcb.15873

38 Scoville, C., **Chapman**, **M.**, Amironesei, R., & Boettiger, C. (2021). Algorithmic conservation in a changing climate. *Current Opinion in Environmental Sustainability*, *51*, 30–35. https://doi.org/10.1016/j.cosust.2021.01.009

39 **Chapman**, **M.**, Walker, W. S., Cook-Patton, S. C., Ellis, P. W., Farina, M., Griscom, B. W., & Baccini, A. (2020). Large climate mitigation potential from adding trees to agricultural lands. *Global Change Biology*, *26*(8), 4357–4365. https://doi.org/10.1111/gcb.15121

40 Griscom, B. W., Busch, J., Cook-Patton, S. C., Ellis, P. W., Funk, J., Leavitt, S. M., Lomax, G., Turner, W. R., **Chapman**, **M.** et al. (2020). National mitigation potential from natural climate solutions in the tropics. *Philosophical Transactions of the Royal Society B*, *375*(1794), 20190126. https://doi.org/10.1098/rstb.2019.0126

41 Oestreich, W. K., **Chapman**, **M.**, & Crowder, L. B. (2020). A comparative analysis of dynamic management in marine and terrestrial systems. *Frontiers in Ecology and the Environment*, *18*(9), 496–504. https://doi.org/10.1002/fee.2243

42 Samndong, R. A., Bush, G., Vatn, A., & **Chapman**, **M.** (2018). Institutional analysis of causes of deforestation in redd+ pilot sites in the equateur province: Implication for redd+

in the democratic republic of congo. *Land Use Policy*, *76*, 664–674. https://doi.org/10.1016/j.landusepol.2018.02.048

43 Cunningham, C., Chen, W. C., Shorten, A., McClurkin, M., Choezom, T., Schmidt, C. P., Chu, V., Bozik, A., Best, C., **Chapman**, **M.** et al. (2014). Impaired consciousness in partial seizures is bimodally distributed. *Neurology*, *82*(19), 1736–1744. https://doi.org/10.12122FWNL.0000000000000404

44 Galvin, B. D., Li, Z., Villemaine, E., Poole, C. B., **Chapman**, **M.**, Pollastri, M. P., Wyatt, P. G., & Carlow, C. K. (2014). A target repurposing approach identifies n-myristoyltransferase as a new candidate drug target in filarial nematodes. *PLoS neglected tropical diseases*, *8*(9), e3145. https://doi.org/10.1371/journal.pntd.0003145

Fellowships and Grants

2026-2030 **EU Horizon Grant** "Biodiversity methods for advanced monitoring at large scales (BEAGLE)" (~$11 million EUR) WP Lead: "Developing impactful insights for the science-policy interface" ($437,000 EUR)

2026-2027 **IVADO "Thematic Semester" Grant (based at McGill University)** "Biodiversity forecasting and planning with AI" Organizing Committee Member (total grant: ~$220,000 CHF)

2024-2027 **NASA - Biological Diversity and Ecological Conservation** Co-PI (total grant: $620,000)

2023 **NCEAS Director's Postdoc Fellowship**, National Center for Ecological Analysis and Synthesis (~$140,000)

Peccei Award, International Institute of Applied Systems Analysis (IIASA) (approx. $7,000)

Moore Foundation, (research funding written into larger grant) (approx. $7,000)

Departmental Research Fellowship, University of California Berkeley ($17,000)

2022 **International Institute of Applied Systems Analysis (IIASA) Summer Fellowship**, Funded through the National Academy of Science ($7,000)

Data Science Teaching Fellowship, Funded through the UC Berkeley Social Science Data-Lab ($5,000)

Artificial Intelligence, Ethics, and Society (AIES-22) Conference Student Award, Funded through the National Science Foundation ($1500)

Environmental Data Science Summit travel grant ($800), NCEAS (delayed to 2023 due to COVID)

2021 **SESYNC Graduate Student Pursuit: Co- PI** (project link) (approx. $35,000)

2020 **Berkeley Center For Technology, Society, and Policy Fellowship** (project link) ($4,000)

2018 **NSF National Research Traineeship** Environment and Society: Data sciences for the 21st Century ($32,000)

NSF Graduate Research Fellowship Program Honorable Mention.

2014 **Foreign Language Area Studies (FLAS) Fellowship: Kiswahili** ($35,000 over two awards)

Selected Talks

2026
- **Chapman, M**. Digital technology refusal in environmental science, policy, and practice *D-USYS Department Retreat, Invited talk*
- **Chapman, M**. Technology refusal in environmental governance *Nordic Environmental Social Science Conference (NESS), Workshop presentation*
- **Chapman, M**. Biosphere governance in an age of algorithms *ETH Zurich, Inaugural Lecture*
- **Chapman, M**. Taking the pulse of the planet *ETH Global Lecture Series, Invited Lecture*
- **Chapman, M**. Towards a just planetary future in the age of AI *Climate Informatics Conference, Invited Keynote*
- **Chapman, M**. tech refusal D-USYS conference
- **Chapman, M**. From international agreements to individual actions: Modeling cooperative action across scales in environmental management *Environmental Informatics Hub, Invited Seminar*
- **Chapman, M**. Can AI help us make more informed restoration and conservation decisions? *Centro Euro-Mediterraneo sui Cambiamenti Climatici (CMCC) Annual Conference, Invited Keynote*

2025
- **Chapman, M**. From data to decisions: Devising equitable strategies for meeting multilateral conservation and restoration targets. *Institute for Environmental Decisions (IED) seminar, Invited Talk*
- **Chapman, M**. Meeting biodiversity targets under future land use demands *University of Bern, Invited Seminar*
- **Chapman, M**. The colonial infrastructures of contemporary biodiversity data. *Living Data Conference, Bogota*
- **Chapman, M**. Towards strategic and decision-informed biodiversity monitoring *Living Data Conference, Bogota*
- **Chapman, M**. Just urban biodiversity futures *Urban Biosphere Futures Meeting, Chicago. Invited talk.*
- **Chapman, M**. From data to decisions: Devising equitable strategies for meeting global biodiversity targets *AIxEnvironment Meeting, Invited Speaker*
- **Chapman, M**. Building a Resilient Future in a Changing Climate *Organization for Security and Co-operation in Europe (OSCE), Invited Panelist*
- **Chapman, M**. Designing financial mechanisms to support targeted conservation and restoration *Impact Evaluation and Mechanism Design Workshop, Iceland*
- **Chapman, M**. Meeting biodiversity targets under future land use demands. *Cumo*
- **Chapman, M**. Harnessing AI: safeguarding high-integrity data for climate action *London School of Economics and Public Policy, Invited Panelist*
- **Chapman, M**. From data to decisions: Towards a just planetary future in the age of artificial intelligence. *Cambridge Center for Human Inspired AI (CHIA) Conference, Invited Keynote*
- **Chapman, M**. Designing coordinated land use strategies. *AI for Climate Cooperative Action Coordination Workshop (Lausanne, CH), Invited Talk*
- **Chapman, M**. Meeting biodiversity targets under changing land use demands *Measuring Development World Bank, Invited Talk*

2024
- **Chapman, M**. Can AI help us make more informed conservation decisions? *Aspen Global Change Institute (AGCI), AI and Biodiversity Workshop, Invited Talk*

Selected Talks (continued)

- **Chapman, M**. Addressing the social and political dimensions of biodiversity data. *North American Congress for Conservation Biology (NACCB), Vancouver 2024*
- **Chapman, M & Faxon, H.**. The colonial infrastructures of contemporary biodiversity data. *Data Justice Workshop, Barcelona 2024*
- **Chapman, M & Faxon, H.**. The colonial infrastructures of contemporary biodiversity data. *World Biodiversity Forum, Davos 2024*
- **Chapman, M**. Do biodiversity data biases risk entrenching social inequity into policy strategies? *Resources for the Future (Invited talk)*
- **Chapman, M**. From individual decisions to international agreements: addressing biodiversity loss in the age of algorithms. *Oregon State University (Invited Seminar)*
- **Chapman, M**. From individual decisions to international agreements: addressing biodiversity loss in the age of algorithms. *National Center for Ecological Analysis and Synthesis (Invited Roundtable Seminar)*
- **Chapman, M**. From data to decisions: toward a just and sustainable planetary future in the age of AI. *University of Oregon (Invited Seminar)*
- **Chapman, M**. From individual decisions to international agreements: addressing biodiversity loss in the age of algorithms. *Stony Brook University (Invited Seminar)*
- **Chapman, M**. AI is transforming the way we confront climate change and biodiversity loss. *GreenBiz Sustainability Conference (Invited keynote panel)*

2023
- **Chapman, M**. AI for equitable climate action. *GLOCAL Innovation Grant Kick-off (invited keynote speaker)*
- **Chapman, M**. Human histories shape the biodiversity data that decide our future. *GEO BON Conference, Montreal 2023)*
- **Chapman, M**. From individual decisions to international agreements: Addressing biodiversity loss in an age of algorithms. *University of California Santa Barbara (invited departmental seminar)*
- **Chapman, M**. Addressing biodiversity loss in an age of algorithms. *International Institute for Applied Systems Analysis (invited seminar).*
- **Chapman, M**. From individual decisions to international agreements: Addressing biodiversity loss in an age of algorithms. *University of California Berkeley (Wildlife seminar)*

2022
- **Chapman, M**, Jung, M., and Visconti, P.. Multiscale prioritization of conservation and restoration measures to meet 2030 biodiversity targets in the EU. *IIASA Summer Symposium* [**Slides**]
- **Chapman, M**, Boettiger, C, and Brashares, J. Potential contributions of private lands to U.S. 2030 biodiversity targets. *ESA 2022* [**Slides**]
- **Chapman, M**. Climate mitigation and biodiversity contributions of land conservation and management (as part of a panel on "Ecologists Perspectives on COP26") *ESA 2022*. [**Slides**]
- **Chapman, M**. Governing AI Applications To Monitoring and Managing Our Global Environmental Commons. *AAAI/ACM conference on Artificial Intelligence, Ethics, and Society (AIES 2022)*. [**Slides**]

2021
- **Chapman, M.**, Schell, C., Brashares, J. "30x30: The New Conservation". Breakthroughs Magazine Virtual Series. [**Recording**]
- **Chapman, M.**. Pathways to 30x30: Accelerating Conservation of California's Nature. California Biodiversity Network Bioinformatics and Conservation Planning round table.

Selected Talks (continued)

- **Chapman, M.**, Boettiger, C. From data to decisions: Algorithms, power, and effective ocean management. UN FAO global forum on AI for a digital blue Planet. [**Recording**]

2020
- **Chapman, M.**. Large climate mitigation from adding trees to agricultural lands. The Nature Conservancy Seminar Series (Invited Talk).
- **Chapman, M.**. Large climate mitigation from adding trees to agricultural lands. Woodwell Climate Research Center Friday Seminar Series (Invited Talk).
- **Chapman, M.**, et al. Tipping points in diversified farming systems. Ecological Society of America 2020 Meeting. Contributed Talk. [**Recording**]

2018
- **Chapman, M.**, and Walker, W. (2018). A Global Analysis of Woody Aboveground Carbon Storage in Crop and Pasture lands. AGU Fall Meeting. (Presentation)

Working Groups and Assessments

2026
- Decision Sovereignty for Nature, Governing and Funding the Commons in an AI Era *Organizer*

2025-2027
- IPBES Spatial Planning and Connectivity Assessment (Review editor)

2024-2025
- USGS North American Biodiversity and Climate Assessment (Chapter Lead Author)
- NSF/NCEAS Biodiversity Data & Conservation Science working group

2024
- Data Justice Workshop (Invited paper/presentation)
- Resources for the Future: Climate Change and Natural Capital Workshop

2022-2024
- Ethics and Practices of Algorithmic Conservation Reading Group **(link)** *Co-founder/organizer*

2023
- Environmental Data Science Innovation Summit (ESIIL)

2023, 2024
- Environmental Data Science Summit (NCEAS)

2022
- AI-Assisted Decision-Making for Conservation (Harvard Center for Research on Computing and Society)

2019-2021
- Ecological Forecasting Initiative *Student Working Group Co-chair and Co-founder*

2021
- UC Berkeley Data and Environment Working Group *Co-founder*
- Bioinformatics and Community Science Round Table steering committee, California Biodiversity Network
- Culturally Relevant Education in Environmental Data Science (CREEDS) Workshop

2020
- SESYNC Cyberinfrastructure Summer Institute
- NIMBioS Adaptive Management Tutorial
- People, Land, and Ecosystems: Leveraging NEON for Socio-Environmental Synthesis

2019
- National Ecological Observation Network (NEON) Science Summit
- Advancing Integrated Process-Based Modeling of Socio-Environmental Systems (SESYNC)
- Graduate Student Workshop on Socio-Environmental Synthesis (SESYNC)
- Ecological Forecasting Initiative Summer Course

Working Groups and Assessments (continued)

- 2017 — Mathematical Ecology Working Group: Woods Hole, MA

Teaching and Mentoring

- PhD Students
  - Jory Griffith (2026–present)
  - Eduardo Gaggi (2026–present)
  - Elodie Adam (2025-present)
  - Giacomo Degaldo (2025-present)
  - Nial Perry (2025-present)
- Master's Students
  - Lydia Dai (2026); Muchen Wang (2026-present); Alessandra Ciampi (2026-present)
- ETH Courses
  - Big Data for Public Policy (Spring 2025)
  - Data Science for Environmental Policy (Fall 2026; course website)
  - Quantitative Policy Analysis and Modeling (Fall 2025, Fall 2026)
  - Topics in Plant Science (Fall 2025)
- ETH Lectures
  - Forest Management (Forest management policy) (2025)
- Co-organizer
  - Science, Policy & Global Crises: Navigating Risk and Uncertainty (Summer School Faculty Mentor, 2026)
  - Climate Change AI In-Person Summer School (2023)
  - Climate Change AI Virtual Summer School (2023, 2024)
- Data Science Instructor
  - UC Berkeley Social Science Data Lab; (1) Data wrangling (2) Deep learning in Python (3) Introduction to R, and (4) Data visualization (2022-2023)
- Graduate Student Instructor
  - UC Berkeley; ESPM 157: Data Science for Global Change Ecology (2020)
  - UC Berkeley; Fung Fellowship Conservation and Technology Course (2022)
- Guest Lectures
  - UC Santa Barbara; Data Science (2023)
  - Stanford University; Introduction to conservation planning and practice (2023)
  - University of California Santa Barbara; Introduction to Remote Sensing (2023)
  - Stanford University; Introduction to conservation planning and practice (2022)
  - Trinity College; U.S. Environmental Policy, Partisanship, and the Global Climate Crisis (2022)
  - UC Berkeley; Conservation and Technology (2022)
  - Middlebury Institute of International Studies; International Marine Science and Policy (2022)
  - Middlebury Institute of International Studies; Ecological Analysis (2022)

Teaching and Mentoring (continued)

Research Mentor	Undergraduate Research Apprentice Program (URAP) (2020-2022)
	Undergraduate Honors Thesis Program (2019-2022)
Technical Mentor	IPAM; Public Policy Course (2017)
Undergraduate Instructor	Yale University; Physics I (2014)
	Yale University; Organic Chemistry II (2013)

Policy Documents and Briefs

2026 **California's 30x30 Initiative: A Subnational Model for Biodiversity Action**, Regions4 Case Study Database **(In press)**

2024 **Land protection and restoration tools to effectively meet 2030 conservation targets in the United States**, USDA Internal Brief [**PDF available upon request**]

2022 **Pathways to 30x30 California: Accelerating Conservation of California's Nature**, Scientific/Technical Writer [**PDF**]

- **Conserving California: Advancing Science in Support of 30x30**, Scientific Writer and Facilitator [**PDF**]
- **California's Pathways to 30x30: Conserving Freshwater Ecosystems**, Legislative Summary; Lead Scientific Writer [**PDF**]
- **California's Pathways to 30x30: Expanding Access to Nature**, Legislative Summary; Contributing Scientific Writer [**PDF available upon request**]
- **California's Pathways to 30x30: Working Lands and Other Effective Conservation Measures (OECMs)**, Legislative Summary; Contributing Scientific Writer [**PDF available upon request**]
- **California's Pathways to 30x30: Partnering with California Native American Tribes**, Legislative Summary; Contributing Scientific Writer [**PDF available upon request**]

2021 **Advancing 30x30 and Protecting Biodiversity**, Lead Scientific Writer [**PDF**]

2018 **Prioritizing Areas for Reforestation of Private Lands in the Brazilian Amazon**. Policy Brief. [**PDF**]

2016 **Analysis of National Circumstances in the Context of REDD+ and Identification of REDD+ Abatement Levers in Papua New Guinea** Report produced by the Wildlife Conservation Society. [**PDF**]

Professional Service and Outreach

2026	Mistra BIOPATH Grants Evaluation Committee
2026-2026	D-USYS Diversity Committee Faculty Member
2025	AI for Public Goods Fast Grants (AI4PG) Committee Member
2024	Climate Change AI Innovation Grants Chair
	Session organizer at World Biodiversity Forum
	Session organizer at North American Convention for Conservation Biology
2023-2024	Climate Change AI Core Team
2022	Graduate Programs Committee student representative (ESPM, UC Berkeley)

Professional Service and Outreach (continued)

2021-2022 Graduate Admission Committee student representative (ESPM, UC Berkeley)

2018-2021 UC Berkeley Graduate Student Association (GSA)

2019-2021 Letters to a Pre-scientist: *Volunteer*

2018-2021 Bay Area Scientists in Schools (BASIS): *Instructor*